

Supplementary Material

A Controlled Benchmark for Statistical Detection of Noise-Induced Transitions in Nonlinear Stochastic Dynamics

Table 1: Supplementary Table S1. Bootstrap confidence intervals for mean F1, computed by resampling independent test series.

Scenario	Method	Mean F1 [95% CI]
D=0.5, white	SNHT	0.725 [0.650, 0.800]
D=0.5, white	Chow	0.645 [0.555, 0.733]
D=0.5, white	GRU	0.253 [0.182, 0.329]
D=0.5, white	CatBoost	0.243 [0.175, 0.313]
D=0.5, white	LSTM	0.207 [0.144, 0.273]
D=1, white	Transformer	0.698 [0.682, 0.714]
D=1, white	LSTM	0.675 [0.652, 0.696]
D=1, white	GRU	0.651 [0.630, 0.670]
D=1, white	CUSUM	0.534 [0.518, 0.551]
D=1, white	CatBoost	0.479 [0.469, 0.490]
D=1, pink	Transformer	0.566 [0.555, 0.577]
D=1, pink	GRU	0.551 [0.541, 0.561]
D=1, pink	LSTM	0.520 [0.508, 0.530]
D=1, pink	CUSUM	0.402 [0.391, 0.413]
D=1, pink	CatBoost	0.216 [0.209, 0.222]
D=1.5, white	Transformer	0.643 [0.631, 0.655]
D=1.5, white	LSTM	0.599 [0.586, 0.612]
D=1.5, white	CUSUM	0.595 [0.586, 0.603]
D=1.5, white	GRU	0.429 [0.412, 0.446]
D=1.5, white	CatBoost	0.341 [0.334, 0.348]

Event-detection strength and localization error at margin 25

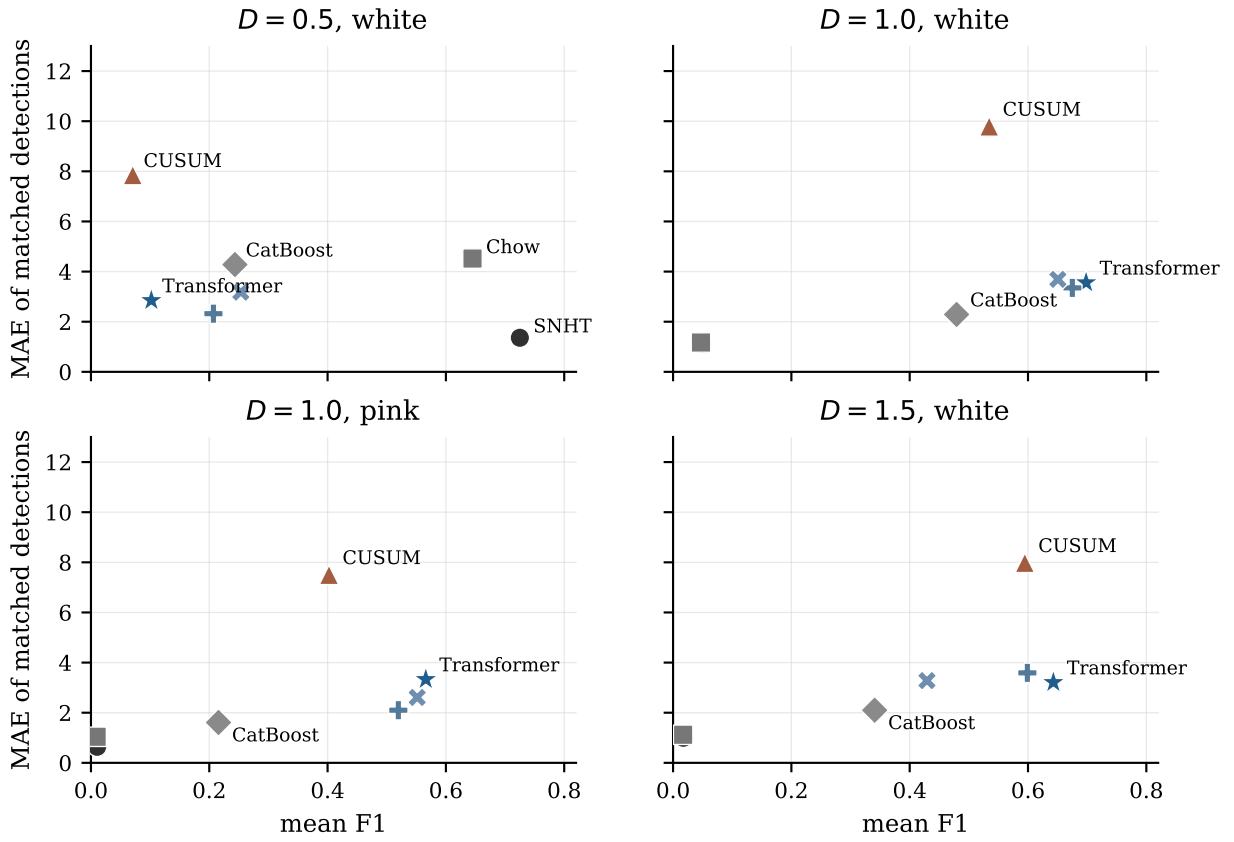


Figure 1: Supplementary Fig. S1. Mean F1 and mean absolute localization error for selected methods at the primary matching margin. MAE is computed only for matched detections and should be read together with F1.